6.1 Polynomials Whose Roots Are Roots of Unity

WORKSHEET

The question

STEP III 2010, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Every nth root of unity satisfies $z^n = 1$. But the smallest positive power that gives 1 might be less than n. For each of the four 4th roots of unity, 1, i, -1, $-\mathrm{i}$, what is that smallest power?

STAGE 1 The first few

Ask yourself Which roots of unity have order 1, 2, 3, 5 and 6? (Draw the 6th roots of unity and sort them by order.)

STAGE 4 No cyclotomic polynomial is a product of two others

Pause and try Suppose $\mathrm{C}_q = \mathrm{C}_r\mathrm{C}_s$. Take a root of C_r. What is its order? What does the identity say about its order? What does that force?

What's really going on

Factorising $x^n - 1$

Every nth root of unity has some order d that divides n, and every dth root of unity of order d is an nth root of unity when $d \mid n$. So the roots of $x^n - 1$ split into classes by order, giving

$$x^n - 1 = \prod_{d \mid n} \mathrm{C}_d(x).$$

For example $x^6 - 1 = \mathrm{C}_1\mathrm{C}_2\mathrm{C}_3\mathrm{C}_6 = (x-1)(x+1)(x^2+x+1)(x^2-x+1)$. Figure 6.1.1 shows the twelve 12th roots of unity sorted in this way. The number of primitive nth roots, which is the degree of C_n, is the number of k between 1 and n with no common factor with n: *Euler's totient function* $\varphi(n)$. Comparing degrees gives the identity $\sum_{d \mid n} \varphi(d) = n$.

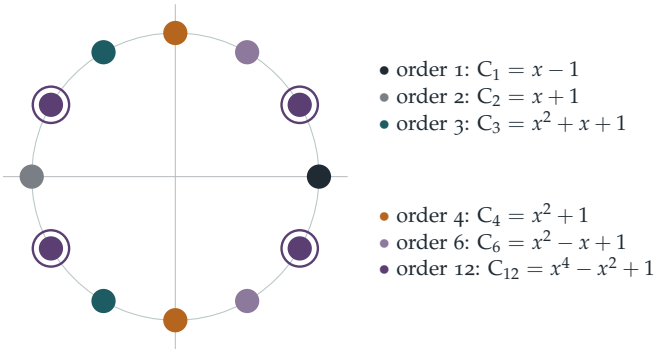


Figure 6.1.1 The twelve 12th roots of unity, coloured by order. The four primitive 12th roots (ringed) are the roots of C_{12}. Together the classes give $x^{12} - 1 = \mathrm{C}_1\mathrm{C}_2\mathrm{C}_3\mathrm{C}_4\mathrm{C}_6\mathrm{C}_{12}$, with degrees $1 + 1 + 2 + 2 + 2 + 4 = 12$.

Integer coefficients, and a surprise

It is not obvious from the definition, but every cyclotomic polynomial has *integer* coefficients. (Use the factorisation of $x^n - 1$ and induction: C_n is $x^n - 1$ divided by a product of earlier cyclotomic polynomials, and long division of integer polynomials by monic integer polynomials keeps the coefficients integers.) For a long time all known examples had coefficients 0, 1 and -1. The first exception is C_{105}, which contains the term $-2x^7$, found by Migotti in 1883; $105 = 3 \times 5 \times 7$ is the smallest number with three distinct odd prime factors. In fact the coefficients of cyclotomic polynomials can be made as large as you like.

Irreducibility

Part (iv) shows that no cyclotomic polynomial is a product of two others. Something much stronger is true: C_n cannot be factorised at all into polynomials with rational coefficients of lower degree. Gauss proved this for prime n (it follows from *Eisenstein's criterion*, which you will meet in a first algebra course), and Kronecker and Dedekind for general n. This *irreducibility* is what makes the cyclotomic polynomials the "primes" among polynomials related to roots of unity, and it is the starting point for the Galois theory of roots of unity.

Try this next

1. **Check the factorisation.** Verify that $(x-1)(x+1)(x^2+x+1)(x^2-x+1) = x^6-1$, and that $x^{12}-1$ factorises as in Figure 6.1.1.

2. **Twice a prime.** Show that if p is an odd prime then $C_{2p}(x) = C_p(-x)$. Check this for $p = 3$.

3. **Totients.** Show that $\varphi(p^k) = p^k - p^{k-1}$ for a prime p, and that $C_{p^k}(x) = C_p\left(x^{p^{k-1}}\right)$. Use this to find C_9.

4. **A longer one.** HARDER Using $x^{15}-1 = C_1C_3C_5C_{15}$, show that $C_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$.

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). The standard undergraduate introduction; cyclotomic polynomials and the regular polygon appear early.
- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press). Includes roots of unity and their arithmetic.
- **Marcus du Sautoy, *Finding Moonshine*** (Fourth Estate). A popular account of symmetry and group theory, including the story of Galois.

The full guided solution is in Section 6.1:
https://beyondstep.keithmcnulty.org/ch06/cyclotomic.html